

Byeonghun Lee

PhD student

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EDUCATION

Mar. 2017 ~ Feb. 2022	Daegu Gyeongbuk Institute of Science and Technology School of Undergraduate Studies <i>B.S. in Convergence</i> GPA: 3.21 / 4.3	Daegu, Korea
Mar. 2022 ~ Present	Korea University School of Electrical Engineering <i>Advisor: Kyong Hwan Jin</i> <i>Ph.D. Student</i> GPA: 4.38 / 4.5	Seoul, Korea

RESEARCH INTERESTS

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- Image processing
 - Inverse problem
 - Retrieval-Augmented Generation in Vision
 - Image Super-Resolution
 - 3D rendering, Depth Estimation

PUBLICATIONS

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1. **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im†, and K.H. Jin†, “ReAL: Reference-to-Image (R2I) Aware Latent Diffusion for Image Super-Resolution,” European Conference on Computer Vision (ECCV), 2026.
 2. **Byeonghun Lee***, Hyunmin Cho*, Honggyu Choi, Soo Min Kang, Iljun Ahn and K.H. Jin*, “Reference-based Super-Resolution via Image-based Retrieval-Augmented Generation Diffusion,” IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
 3. Woo Kyoung Han*, Yongjun Lee*, **Byeonghun Lee**, Sang Hyun Park, Sunghoon Im*, and K.H. Jin*, “JPEG Processing Neural Operator for Backward-Compatible Coding,” IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
 4. Woo Kyoung Han, **Byeonghun Lee**, Hyunmin Cho, Sunghoon Im*, K.H. Jin*, “Towards Lossless Implicit Neural Representation via Bit Plane Decomposition,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

5. EungGu Kang, **Byeonghun Lee**, Sunghoon Im*, and K. H. Jin*, “BurstM: Deep Burst Multi-scale SR using Fourier Space with Optical Flow,” European Conference on Computer Vision (ECCV), 2024.
6. Minsu Kim, Jaewon Lee, **Byeonghun Lee**, Sunghoon Im and K. H. Jin, “Implicit Neural Image Stitching With Enhanced and Blended Feature Reconstruction,” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024.
7. Woo Kyoung Han, **Byeonghun Lee**, Sang Hyun Park, and K. H. Jin, “ABCD : Arbitrary Bitwise Coefficient for De-quantization,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

Under Review & In preparation

1. **Byeonghun Lee**, Jaedeok Kim, Sunghoon Im and Kyong Hwan Jin, "Target-Oriented Neural MPI Stitching for Novel View Synthesis,".

PATENTS

1. Apparatus and Method of Recovering Image Using Arbitrary Bitwise Coefficient Estimation for De-quantization, 102023011027(2023-08-24)
2. Method and Apparatus for Lossless Implicit Neural Representation using Bit-plane Decomposition, 1020250151831(2025-10-20)
3. Polarization-related patents (**Samsung Electronics collaboration**)
- 1 international patent application, pending

PROJECTS

- Synthetic Polarization Data Generation from RGB Images & Polarization-Based Precise Depth Estimation – **Samsung Electronics, GTR** (MAR 2026 –)
- Hallucination Mitigation in Video Understanding Models – **Samsung Research** (DEC 2025 –)
- User-Preference-Based Image Quality Enhancement – **Samsung Research** (Jun 2024 – May 2025)
- Cognitive Reverse-Haptics Reproduction System Lab – **Basic Research Laboratory (BRL), National Research Foundation of Korea** (Feb 2023 – Feb 2024) Developed signal-processing algorithms to convert real-world tactile data into digital signals, and designed a Verilog-based hardware demonstration.
- Food Classification Algorithm for Caloric Intake Analysis – **Pinset Healthcare** (Aug 2022 – Jul 2023) Developed food classification and calorie estimation algorithms from user-captured food images.
- Deep Learning Algorithm for Bacteria Detection – **THE WAVE TALK** (Mar 2022 – Aug 2022) Developed an image-based classification algorithm to identify bacteria species and concentration levels.

AWARDS AND HONORS

- Bronze medal for paper award , IPIU, Korea (Jun. 2026)

SKILLS AND TECHNIQUES

- Proficient in C++ and Python
- Experienced in developing AI algorithms with PyTorch
- Background in 3D rendering and image warping
- Expertise in image super-resolution techniques

- Research and development experience with diffusion models